



UNIVERSITY OF ASIA PACIFIC

**Department of Computer Science and Engineering
(CSE)**

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Experiment Title: Basic Optical Communication System & Configurations.

Submitted by:

Name: MD. Fairuz Anam

Reg ID: 23101122

Section: C-1

Submitted To:

Al-Amin Sany

Lecturer

Department of CSE, UAP

1.Introduction:

The rapid growth of global data traffic has made optical fiber communication an indispensable technology in modern telecommunications. By transmitting information as modulated light pulses through a glass medium, fiber-optic systems offer substantially higher bandwidth, lower signal attenuation, and complete immunity to electromagnetic interference compared to conventional copper-based networks — establishing optical fiber as the preferred medium for long-distance and high-capacity data transmission.

This report documents the design, implementation, and performance evaluation of a complete optical fiber communication system simulated using OptiSystem software. The simulation covers the full end-to-end signal path — from digital data generation at the transmitter, through electro-optic modulation and single-mode fiber propagation, to photodetection and signal recovery at the receiver. Both pre-optimisation and post-optimisation configurations are analysed, enabling direct comparison of system performance under degraded and ideal operating conditions.

The system employs Intensity Modulation with Direct Detection (IM-DD) using Non-Return-to-Zero On-Off Keying (NRZ-OOK) modulation — one of the most widely deployed formats in practical fiber-optic systems due to its simplicity and cost-effectiveness. A Continuous Wave laser operating at 193.1 THz in the ITU-T C-band serves as the optical carrier, modulated by a Mach-Zehnder Modulator driven by a pseudo-random binary sequence.

1.1 Objectives

The primary objectives of this simulation study are:

- To design and implement a complete optical fiber communication link in OptiSystem with realistic component parameter configurations.
- To characterise transmitted and received signal waveforms using time-domain visualizers, observing signal integrity under both degraded and optimised conditions.
- To evaluate key performance metrics including Bit Error Rate (BER), Q-Factor, and eye diagram characteristics at the receiver decision point.
- To analyse the optical spectrum at the transmitter and receiver, examining the effect of fiber propagation on spectral shape and signal quality.
- To conduct a power budget analysis using optical and electrical power meters, quantifying signal loss at each stage of the link.
- To compare system performance before and after parameter optimisation, demonstrating the impact of component configuration on overall link reliability.

2. Component List

- 1. PRBS Generator** Type: Pseudo-Random Parameters: Bit Rate = Bit rate (global parameter) Bits/s
- 2. NRZ Pulse Generator** Type: NRZ Electrical Parameters: Driven by PRBS; outputs NRZ waveform at global bit rate
- 3. CW Laser** Type: Continuous Wave Parameters: Frequency = 193.1 THz, Power = 0 dBm, Linewidth = 10 MHz, Initial Phase = 0°
- 4. Mach-Zehnder Modulator** Type: Electro-optic MZM Parameters: Extinction Ratio = 30 dB, Chirp = Disabled, Symmetry Factor = -1
- 5. Optical Fiber** Type: SMF (G.652) Parameters: Length = 50 km, Attenuation = 0.2 dB/km, Reference Wavelength = 1550 nm
- 6. Photodetector PIN** Type: PIN Diode Parameters: Responsivity = 1 A/W, Dark Current = 10 nA
- 7. Low Pass Butterworth Filter** Type: Butterworth LPF Parameters: Cutoff = $0.75 \times \text{Bit rate Hz}$, Order = 1, Insertion Loss = 0 dB, Depth = 100 dB
- 8. Optical Spectrum Analyzer** Type: OSA (Tx) Parameters: Monitors optical spectrum at MZM output
- 9. Optical Spectrum Analyzer_1** Type: OSA (Rx) Parameters: Monitors optical spectrum after 50 km fiber propagation
- 10. Optical Time Domain Visualizer** Type: OTDV (Tx) Parameters: Monitors optical pulse shape at transmitter output
- 11. Optical Time Domain Visualizer_1** Type: OTDV (Rx) Parameters: Monitors optical pulse shape after fiber propagation
- 12. Optical Power Meter** Type: Power Meter (Tx) Parameters: Measures average optical power at transmitter output
- 13. Optical Power Meter_1** Type: Power Meter (Rx) Parameters: Measures average optical power at fiber output
- 14. Oscilloscope Visualizer** Type: Oscilloscope (Tx) Parameters: Displays NRZ electrical waveform at transmitter
- 15. Oscilloscope Visualizer_1** Type: Oscilloscope (Rx) Parameters: Displays received electrical waveform post-detection before filtering
- 16. Oscilloscope Visualizer_2** Type: Oscilloscope Parameters: Displays received waveform after Low Pass Butterworth Filter
- 17. Electrical Power Meter Visualizer** Type: Electrical Power Meter Parameters: Measures received electrical power before filter

18. Electrical Power Meter Visualizer_1 Type: Electrical Power Meter Parameters: Measures received electrical power after filter

19. BER Analyzer Type: BER / Eye Analyzer Parameters: Computes BER, Q-Factor, and eye diagram at the receiver decision point

3. Working Process

3.1 Overall Architecture

The simulated optical fiber communication system is built upon a three-stage architecture — Transmitter (Tx), Optical Channel, and Receiver (Rx) — which forms the foundation of all practical fiber-optic links. Digital data originates as a pseudo-random binary sequence at the transmitter, where it is shaped into NRZ electrical pulses and used to modulate a continuous-wave optical carrier via a Mach-Zehnder Modulator. The resulting optical signal is then launched into a single-mode fiber span, where it experiences attenuation and chromatic dispersion during propagation, before being recovered at the receiver through photodetection and electrical filtering.

Throughout the link, a comprehensive set of monitoring instruments — optical power meters, spectrum analyzers, oscilloscopes, and time-domain visualizers — are connected at strategic points to capture signal characteristics at each stage. This enables a detailed and systematic evaluation of how signal quality changes from the transmitter output through to the final decision point at the BER Analyzer.

3.2 Transmitter Section

The transmitter converts digital data into a modulated optical signal ready for fiber transmission. Four components work in cascade to achieve this: a PRBS Generator, NRZ Pulse Generator, CW Laser, and Mach-Zehnder Modulator.

3.2.1 Pseudo-Random Bit Sequence (PRBS) Generator

The PRBS Generator serves as the digital data source, producing a statistically random binary stream that closely resembles real-world data traffic. The bit rate is not hardcoded into the component but instead defined as a global simulation parameter, enabling the entire layout to be tested at different data rates through parametric sweeping without requiring individual component modifications. The balanced distribution of 1s and 0s with frequent bit transitions ensures that all performance metrics — including eye diagram quality, BER, and timing recovery — are evaluated under realistic and representative operating conditions.

3.2.2 NRZ Pulse Generator

The NRZ Pulse Generator translates the binary PRBS data into Non-Return-to-Zero electrical waveforms suitable for driving the optical modulator. In NRZ format, a logic '1' maintains a constant high voltage for the full duration of the bit period, while a logic '0' holds a constant low voltage — the signal level never drops to zero between consecutive identical bits. This encoding scheme makes efficient use of the available spectral bandwidth, requiring approximately half the bandwidth of Return-to-Zero (RZ) formats for an equivalent data rate. The resulting NRZ waveform is applied directly to the RF drive port of the Mach-Zehnder Modulator.

3.2.3 CW Laser Source

A Continuous Wave laser provides the unmodulated optical carrier for the system, configured with the following parameters:

- Frequency: 193.1 THz (wavelength $\approx$ 1552.5 nm, ITU-T C-band)
- Output Power: 0 dBm (1 mW)
- Linewidth: 10 MHz
- Initial Phase: 0°

Operation in the C-band is chosen because standard single-mode fiber achieves its minimum attenuation of approximately 0.2 dB/km in this wavelength region, making it the preferred band for long-distance optical transmission worldwide. The 0 dBm launch power keeps the system well within the linear fiber propagation regime, ensuring that nonlinear phenomena such as Self-Phase Modulation remain negligible throughout the link.

3.2.4 Mach-Zehnder Modulator (MZM)

The MZM is an electro-optic device that transfers the NRZ electrical data onto the CW optical carrier through the Pockels effect. The device divides the input laser light between two waveguide arms; when a drive voltage is applied, it alters the refractive index of one arm relative to the other, introducing a phase difference. Upon recombination at the output coupler, this phase difference produces constructive or destructive optical interference — translating the binary drive signal into corresponding variations in optical intensity. The MZM is configured with the following parameters:

- Extinction Ratio: 30 dB — ensuring very deep suppression of the optical OFF state, minimising the receiver power penalty
- Negative Signal Chirp: Disabled — maintaining chirp-free operation to prevent spectral broadening during modulation
- Symmetry Factor: -1 — implementing a push-pull drive scheme that inherently cancels residual chirp and maximises spectral purity

The use of an external MZM rather than direct laser modulation provides superior extinction ratio, cleaner pulse edges, and negligible frequency chirp — all of which directly translate to better BER performance and greater transmission reach.

3.3 Transmission Channel

3.3.1 Optical Fiber

The optical transmission medium consists of a standard single-mode fiber span (SSMF, ITU-T G.652). Three key physical effects govern signal propagation through the fiber:

- **Attenuation:** A loss coefficient of 0.2 dB/km produces approximately 10 dB of total insertion loss over 50 km, reducing the received optical power to roughly -10 dBm at the detector input.
- **Chromatic Dispersion:** The fiber dispersion coefficient of approximately 17 ps/(nm·km) in the C-band causes different spectral components of the optical pulse to arrive at the receiver at slightly different times, producing pulse broadening. Over 50 km, the cumulative dispersion reaches approximately 850 ps/nm, contributing to intersymbol interference at higher bit rates.
- **Nonlinear Effects:** At the 0 dBm launch power used in this simulation, fiber nonlinearities including Self-Phase Modulation and Cross-Phase Modulation remain negligible and do not contribute measurably to signal degradation.

3.4 Receiver Section

The receiver section is responsible for converting the attenuated and dispersed optical signal back into electrical form and conditioning it for bit-error evaluation. The receiver chain in this simulation consists of a PIN Photodetector and a Low Pass Butterworth Filter, with the output fed directly into the BER Analyzer.

3.4.1 PIN Photodetector

The PIN photodiode carries out direct optical-to-electrical conversion by generating a photocurrent that is proportional to the instantaneous received optical power. The device is configured with:

- **Responsivity:** 1 A/W — ideal unity responsivity where each milliwatt of received optical power produces 1 mA of photocurrent
- **Dark Current:** 10 nA — a small leakage current present even in the absence of optical input, contributing to the receiver noise floor

At the received power level of approximately -10 dBm after 50 km of fiber propagation, a standard PIN receiver operates comfortably within its linear range without requiring the internal multiplication gain of an avalanche photodiode.

3.4.2 Low Pass Butterworth Filter

The electrical signal from the photodetector passes through a Low Pass Butterworth Filter before reaching the BER Analyzer. The filter is configured with:

- Cutoff Frequency: $0.75 \times \text{Bit rate Hz}$ (global parameter, automatically scales with bit rate)
- Filter Order: 1
- Insertion Loss: 0 dB
- Depth: 100 dB

The Butterworth filter is selected for its maximally flat amplitude response within the passband, which avoids signal distortion while providing effective rejection of out-of-band noise. The $0.75 \times \text{Bit rate}$ cutoff frequency represents an optimised design choice that maximises noise rejection without introducing excessive intersymbol interference — a well-established trade-off in optical receiver design. The conditioned electrical signal is then passed to the BER Analyzer for final performance assessment.

3.5 System Block Diagram

The figure below shows the complete OptiSystem simulation layout with all transmitter, channel, receiver, and monitoring components as configured in this project:

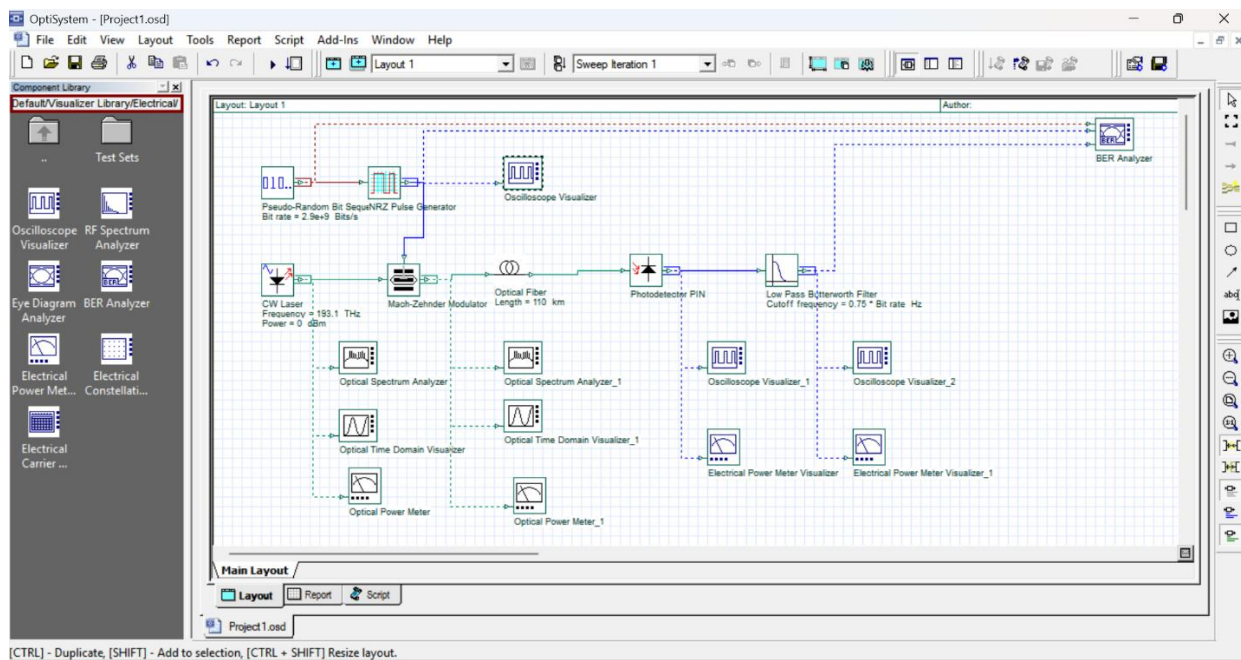


Figure 1: OptiSystem Simulation Layout — Complete Optical Fiber Communication System

4.Results & Discussion

4.1 Signal Waveform Analysis (Tx & Rx)

4.1.1 Transmitted Signal (Tx)

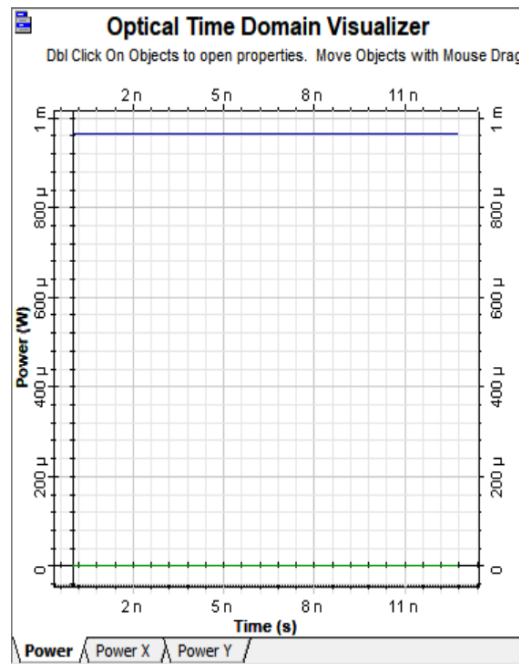


Figure 2.1: Optical Time Domain Visualizer — Transmitted Signal Before Optimization (Flat Line, No Pulse Pattern Visible).

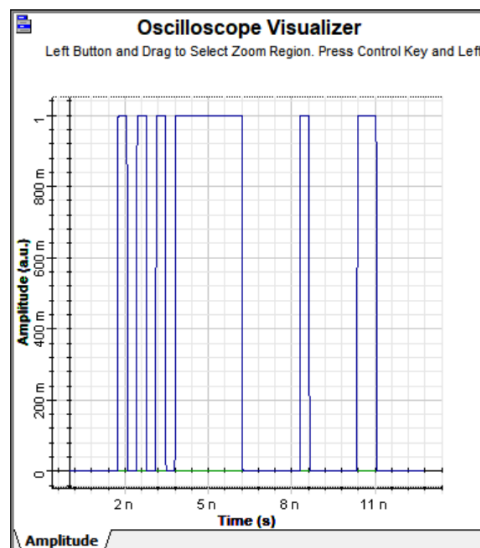


Figure 2.2: Oscilloscope Visualizer — Transmitted NRZ-OOK Electrical Signal After Optimization

Before Optimization:

Figure 2.1 shows the Optical Time Domain Visualizer before optimization. The display shows a completely flat line with no visible pulse transitions, indicating total signal suppression due to the degraded system parameters. No NRZ data pattern is observable at the transmitter output, confirming that the system is non-functional under these conditions.

After Optimization:

Figure 2.2 shows the transmitted electrical signal captured by the Oscilloscope Visualizer at the NRZ Pulse Generator output after restoring the correct parameters. The waveform exhibits ideal NRZ-OOK characteristics — logic '1' bits appear as clean flat-topped pulses at 1 a.u. and logic '0' bits settle precisely at 0 a.u., with sharp noise-free transitions between states. The pseudo-random bit pattern is clearly distinguishable with no visible distortion, overshoot, or noise, representing the undistorted reference waveform before any fiber propagation effects.

4.1.2 Received Signal (Rx)

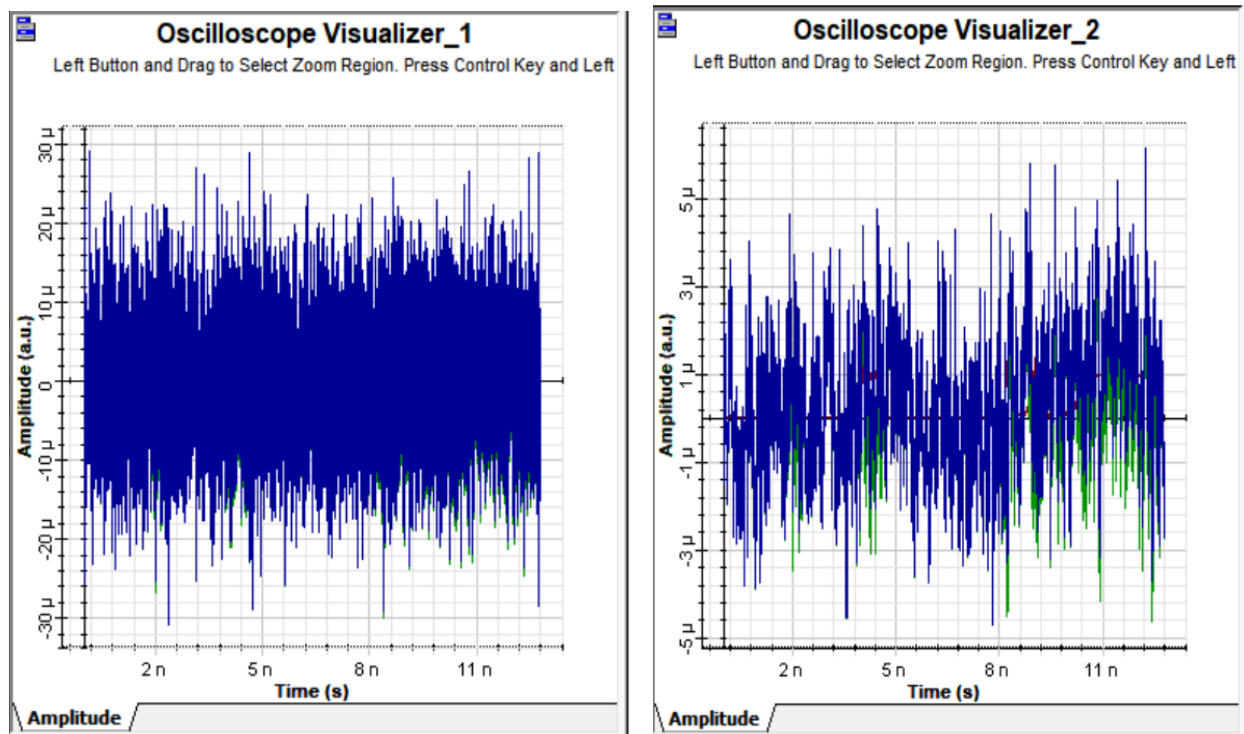


Figure 3.1: Oscilloscope Visualizer_1 — Received Electrical Signal Before Optimization (Completely Noise-Dominated, Signal Indistinguishable).

Figure 3.2: Oscilloscope Visualizer_2 — Filtered Received Signal Before Optimization (Noise Dominant Even After Filtering).

Before Optimization:

Figure 3.1 shows the received electrical signal captured by Oscilloscope Visualizer_1 before optimization. The waveform is completely buried in noise with amplitude reaching only 30–35 μ a.u. The '1' and '0' levels are entirely indistinguishable from each other, making correct bit detection impossible. Figure 3.2 shows the same signal after the Butterworth filter (Oscilloscope Visualizer_2) — even after filtering, the signal remains heavily noise-dominated with amplitude in the range of 5–7 μ a.u., confirming that the filter alone cannot recover a signal this severely degraded.

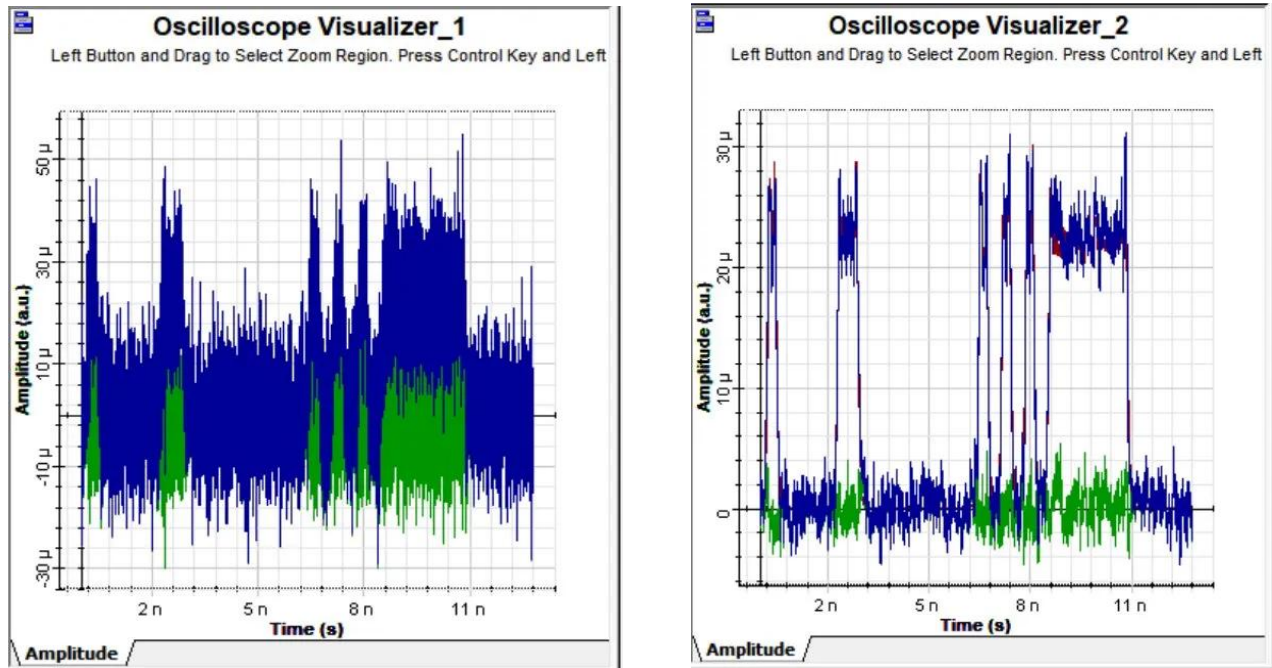


Figure 3.3: Oscilloscope Visualizer_1 — Received Electrical Signal After Optimization (Signal Pattern Visible, '1' Levels At 30–50 μ A.U., 100 Km Fiber Span).

Figure 3.4: Oscilloscope Visualizer_2 — Filtered Received Signal After Optimization (Cleaner '0' Level Noise Floor, Improved SNR After Second-Order Butterworth Filter).

After Optimization:

Figure 3.3 shows the received electrical signal captured by Oscilloscope Visualizer_1 after optimization. The signal amplitude reaches 30–50 μ a.u. for logic '1' levels and a recognisable pulse structure is now clearly visible across the bit sequence, representing a significant improvement over the before case where the signal was completely buried in noise. The '1' level appears as noisy spikes due to PIN detector shot noise and chromatic dispersion accumulated over the 100 km fiber span, while the '0' level (green trace)

carries noise fluctuations in the range of -10 to $10 \mu \text{ a.u.}$ that partially overlap with the lower signal levels.

Figure 3.4 shows the filtered received signal from Oscilloscope Visualizer_2 after the second-order Low Pass Butterworth Filter. The improvement over Figure 3.3 is clearly visible — the '0' level noise floor (green trace) is significantly flatter and more compressed, settling close to $0 \mu \text{ a.u.}$ with much reduced fluctuation. The '1' level pulses now show more defined peaks reaching up to $30 \mu \text{ a.u.}$, with cleaner separation between logic levels. This confirms that the second-order Butterworth filter effectively attenuates high-frequency noise components while preserving the in-band signal, resulting in improved SNR at the BER Analyzer decision point.

4.2 Power Measurements (Tx & Rx)

4.2.1 Transmitted and Received Optical Power

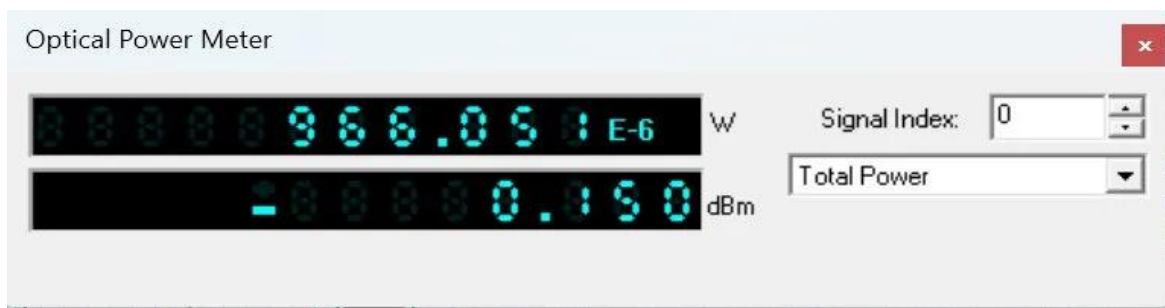


Figure 4.1: Optical Power Meter (Tx) — Transmitted Optical Power Before Optimization:



Figure 4.2: Optical Power Meter_1 (Rx) — Received Optical Power Before Optimization

Before Optimization:

Figure 4.1 shows the Optical Power Meter at the transmitter output reading 966.051×10^{-6} W (-0.050 dBm), and Figure 4.2 shows Optical Power Meter_1 at the fiber output reading 338.501×10^{-6} W (-4.704 dBm). The 4.654 dB difference between the two readings indicates higher than expected fiber loss under the degraded system parameters, reflecting the poor signal transmission conditions before optimization.

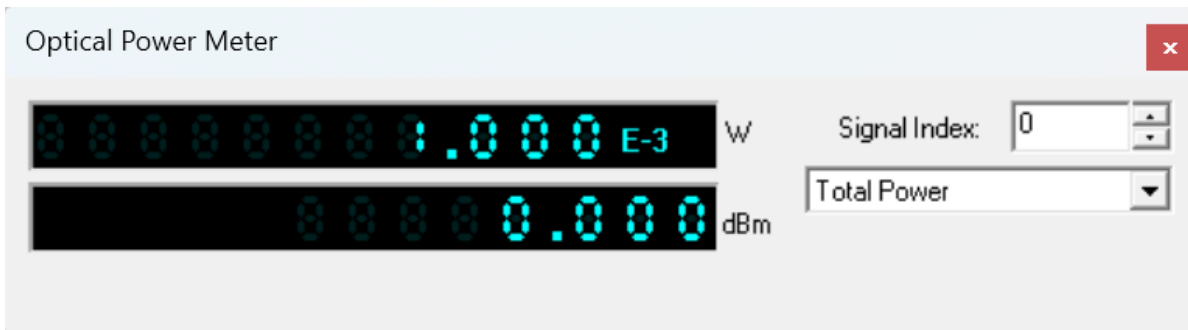


Figure 4.3: Optical Power Meter (Tx) — Transmitted Optical Power After Optimization



Figure 4.4: Optical Power Meter_1 (Rx) — Received Optical Power After Optimization

After Optimization:

Figure 4.3 shows the Optical Power Meter at the transmitter output reading 1.000×10^{-3} W (0.000 dBm), confirming the CW laser is launching the full 0 dBm into the fiber span. Figure 4.4 shows Optical Power Meter_1 at the fiber output reading 350.397×10^{-6} W (-4.554 dBm). The 4.554 dB difference between the two readings reflects the combined effect of fiber attenuation over the 100 km span with 0.15 dB/km attenuation, and the average power reduction due to NRZ-OOK modulation where the signal alternates between ON and OFF states throughout the bit sequence.

4.2.2 Received Electrical Power



Figure 4.5: Electrical Power Meter Visualizer — Received Electrical Power Before Filter, Before Optimization

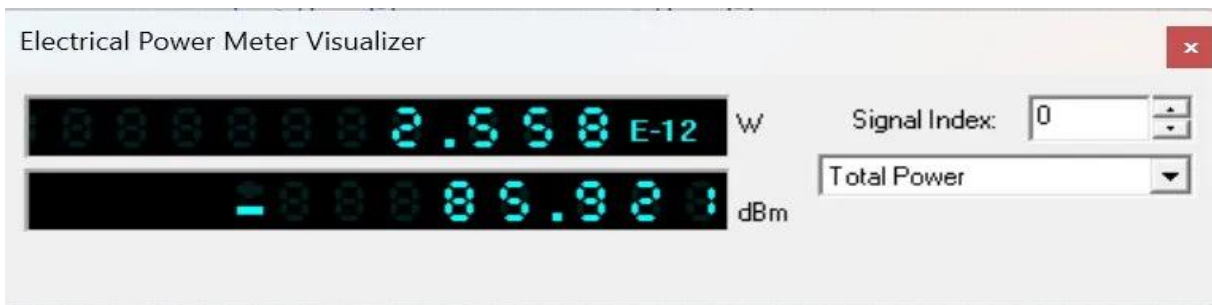


Figure 4.6: Electrical Power Meter Visualizer_1 — Received Electrical Power After Filter, Before Optimization

Before Optimization:

Figure 4.5 shows the Electrical Power Meter Visualizer before filtering reading 63.988×10^{-12} W (-71.938 dBm), and Figure 4.6 shows Electrical Power Meter Visualizer_1 after filtering reading 2.558×10^{-12} W (-85.920 dBm). The large 13.982 dB drop between the two readings indicates that under the degraded system conditions, the filter is aggressively cutting into the signal along with the noise — a sign that the signal power is too weak to be effectively preserved through the filter.

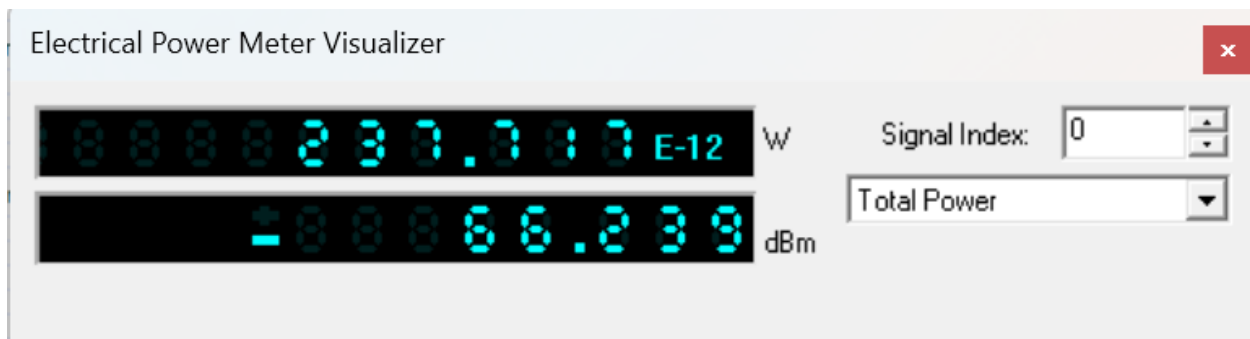


Figure 4.7: Electrical Power Meter Visualizer — Received Electrical Power Before Filter, After Optimization

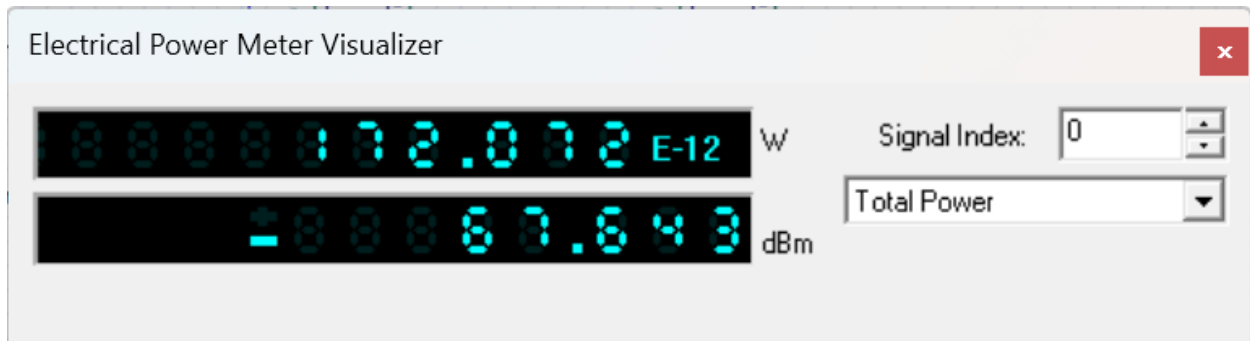


Figure 4.7: Electrical Power Meter Visualizer — Received Electrical Power Before Filter, After Optimization

After Optimization:

Figure 4.7 shows the Electrical Power Meter Visualizer before filtering reading $237.717 \times 10^{-12} \text{ W}$ (-66.239 dBm), representing the total photocurrent power from the PIN photodetector including signal and noise components. Figure 4.8 shows Electrical Power Meter Visualizer_1 after the Butterworth filter reading $172.072 \times 10^{-12} \text{ W}$ (-67.643 dBm). The 1.404 dB reduction between the two readings confirms that the second-order Butterworth filter has successfully attenuated out-of-band noise components while preserving the in-band signal, resulting in improved SNR at the BER Analyzer decision point.

4.3 Optical Spectrum Analyzer (OSA)

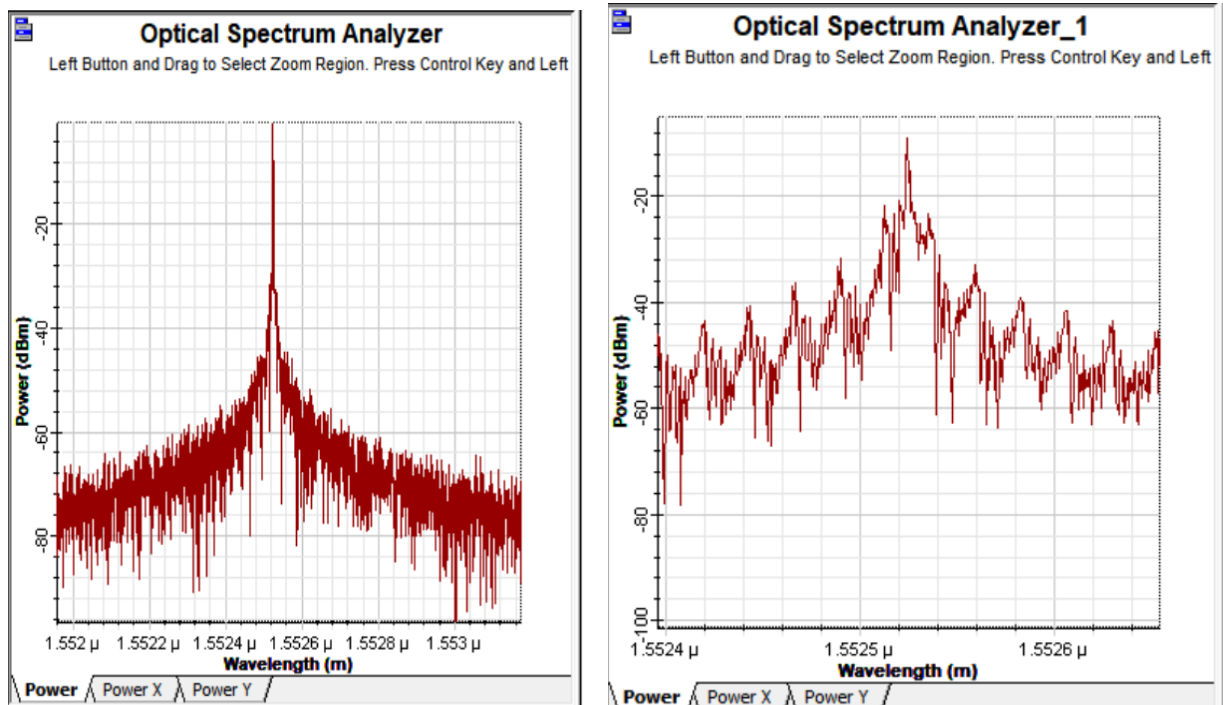


Figure 5.1: Optical Spectrum Analyzer (Tx) — Transmitted Optical Spectrum Before Optimization

Figure 5.2: Optical Spectrum Analyzer_1 (Rx) — Received Optical Spectrum Before Optimization

Before Optimization:

Figure 5.1 shows the transmitted optical spectrum captured by the Optical Spectrum Analyzer before optimization, centred at $1.55253\ \mu\text{m}$ with a peak power of $-0.253\ \text{dBm}$. While a dominant carrier peak is visible, the spectrum exhibits a very wide and heavily noise-laden distribution spanning from $1.5497\ \mu\text{m}$ to $1.5554\ \mu\text{m}$ — a span of approximately $5.7\ \text{nm}$. The noise floor is elevated to approximately $-96.809\ \text{dBm}$, significantly higher than expected for a well-configured system, and the spectral shape lacks the clean sinc-squared NRZ envelope, indicating degraded signal quality.

Figure 5.2 shows the received optical spectrum from Optical Spectrum Analyzer_1 before optimization. The centre wavelength remains at $1.55253\ \mu\text{m}$ but the peak power has already dropped to $-4.503\ \text{dBm}$. The spectrum retains the same wide noisy distribution as the transmitted spectrum, with a noise floor of $-104.547\ \text{dBm}$. The broad spectral spread and irregular shape confirm that the system is operating under poor conditions, with excessive noise contaminating the signal across the entire observable wavelength range.

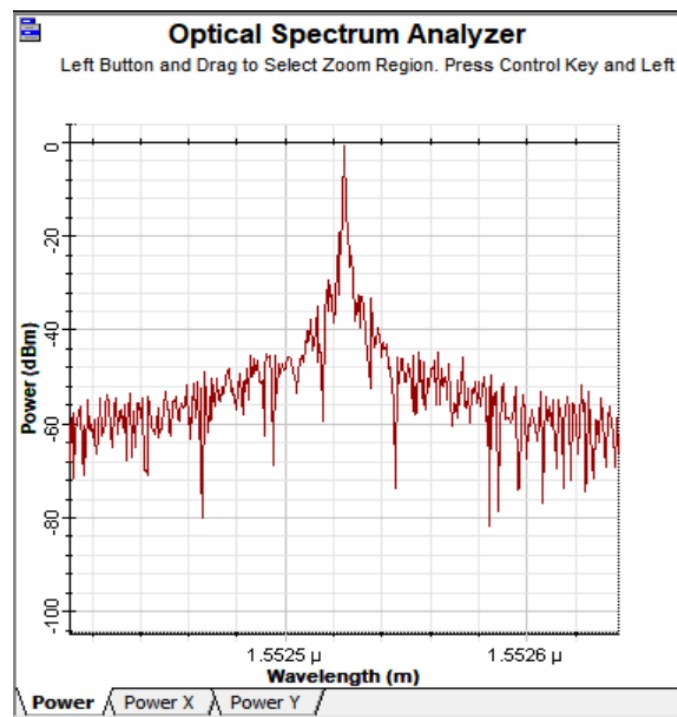


Figure 5.3: Optical Spectrum Analyzer (Tx) — Transmitted Spectrum After Optimization

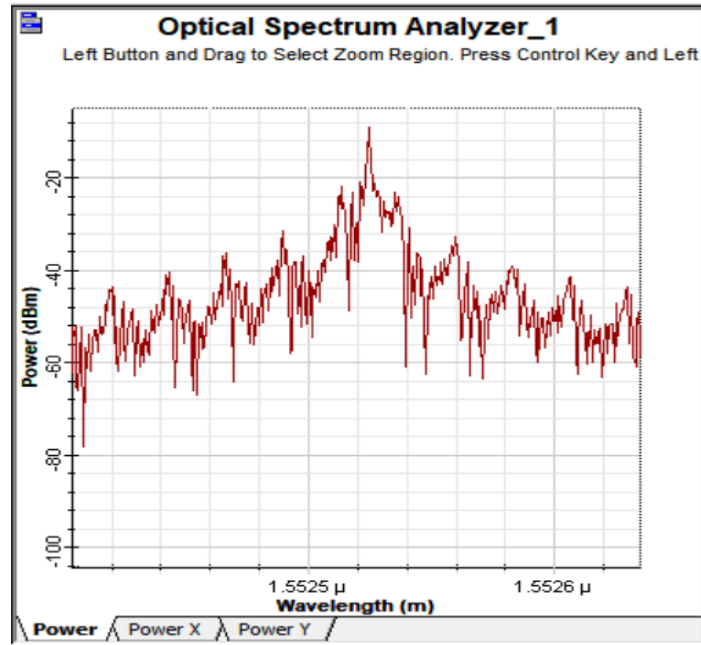


Figure 5.4: Optical Spectrum Analyzer_1 (Rx) — Received Spectrum After Optimization

After Optimization:

Figure 5.3 shows the transmitted optical spectrum after optimization, centred at 1.55252 μm with a sharp clean carrier peak at -0.202 dBm and well-defined NRZ modulation sidebands on either side. Figure 5.4 shows the received optical spectrum from Optical Spectrum Analyzer_1 after 50 km of fiber propagation, with peak power reduced to -3.366 dBm while maintaining the same centre wavelength. The consistent noise floor of approximately -104 dBm in both optimised spectra, combined with the clean spectral shape, confirms linear-regime propagation with no nonlinear distortion throughout the fiber span.

4.4 Eye Diagram Analysis

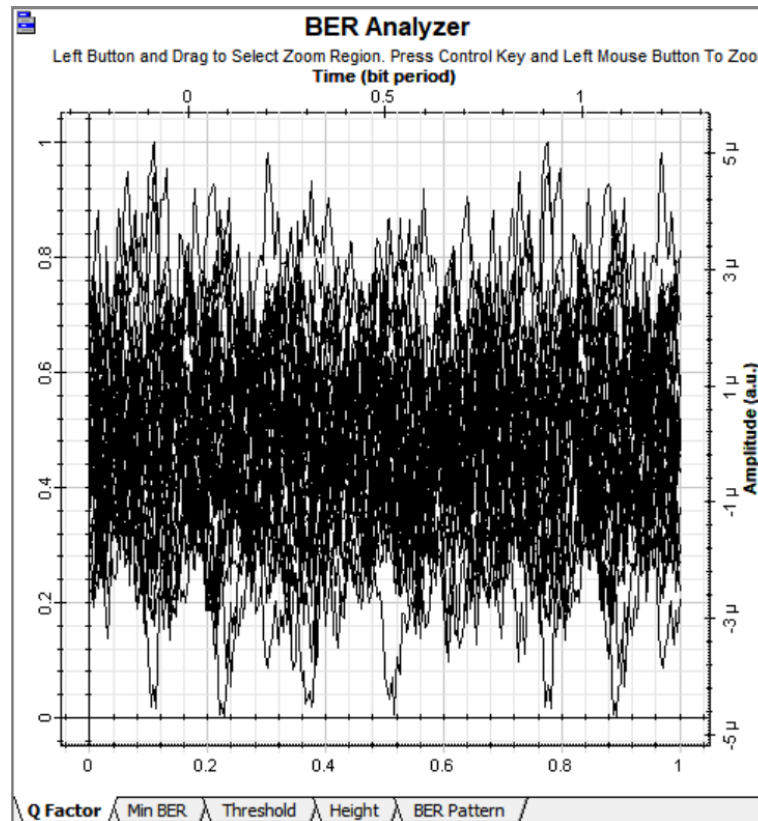


Figure 6.1: BER Analyzer — Eye Diagram Before

Before Optimization:

Figure 6.1 shows the eye diagram from the BER Analyzer before optimization. The eye is completely closed with no discernible opening — the '1' and '0' level traces overlap entirely across the full bit period. The Analysis panel confirms the severity of the degradation: Q Factor = 0, Min BER = 1, Eye Height = 0, Threshold = 0, and Decision Instant = 0. A BER of 1 means every single received bit is incorrect — the system has completely failed under the degraded parameter conditions and no meaningful data recovery is possible.

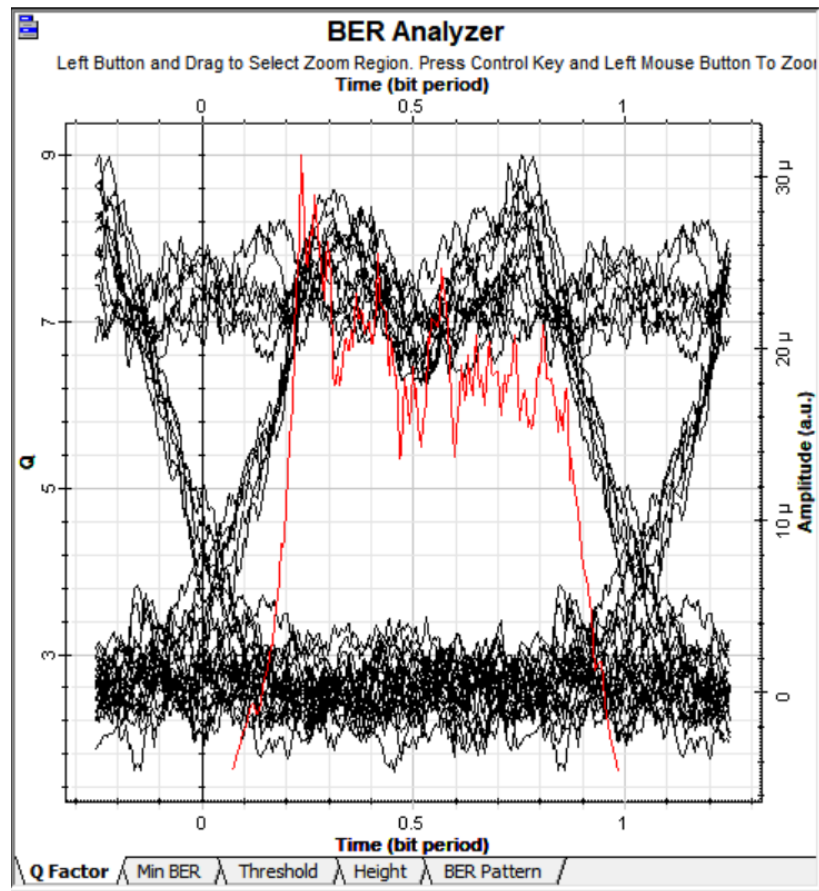


Figure 6.2: **BER Analyzer** — Eye Diagram After

After Optimization:

Figure 6.2 shows the eye diagram after optimization. The eye is clearly open with a measurable eye height of 1.60073×10^{-5} a.u. and a decision threshold of 1.24645×10^{-5} a.u. at a decision instant of 0.23565 bit periods. The Q Factor of 9.00598 and Min BER of 1.04467×10^{-19} represent a dramatic improvement from total system failure to fully error-free operation, confirming that the optimised parameters restore complete signal integrity at the receiver decision point.

4.5 BER and Q Factor Analysis

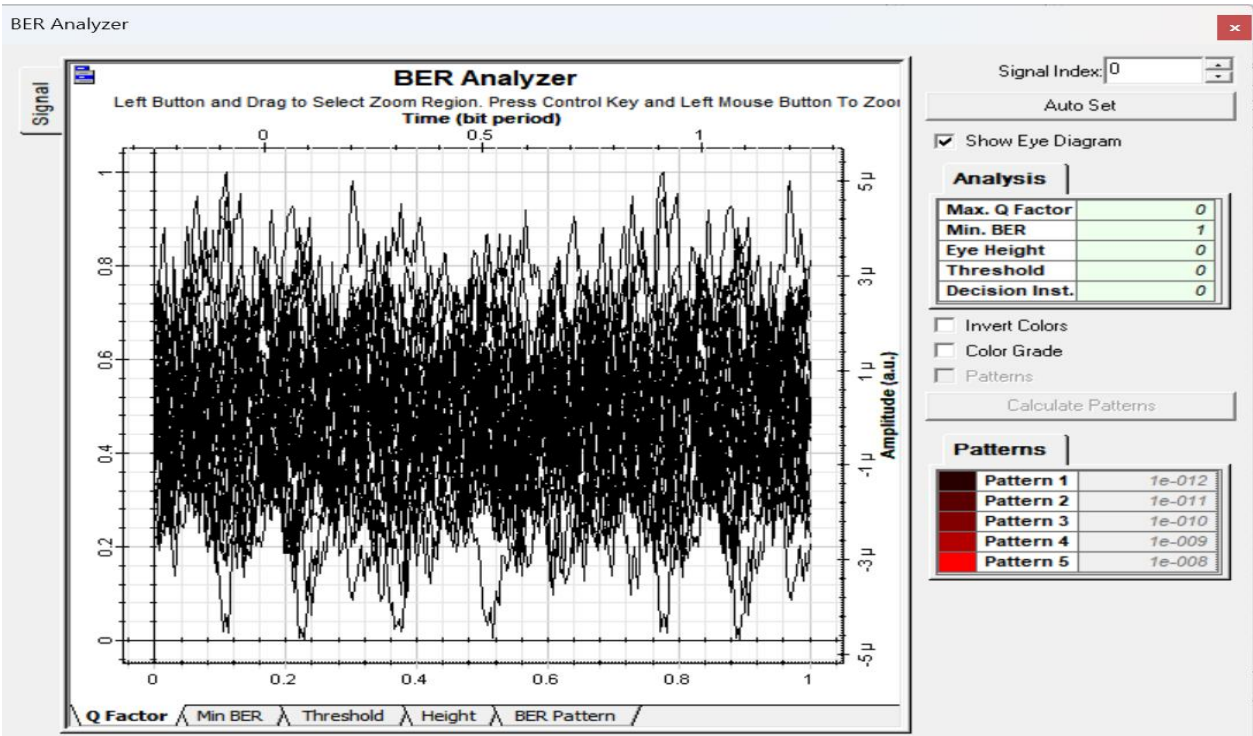


Figure 6.1: BER Analyzer — Eye Diagram Before

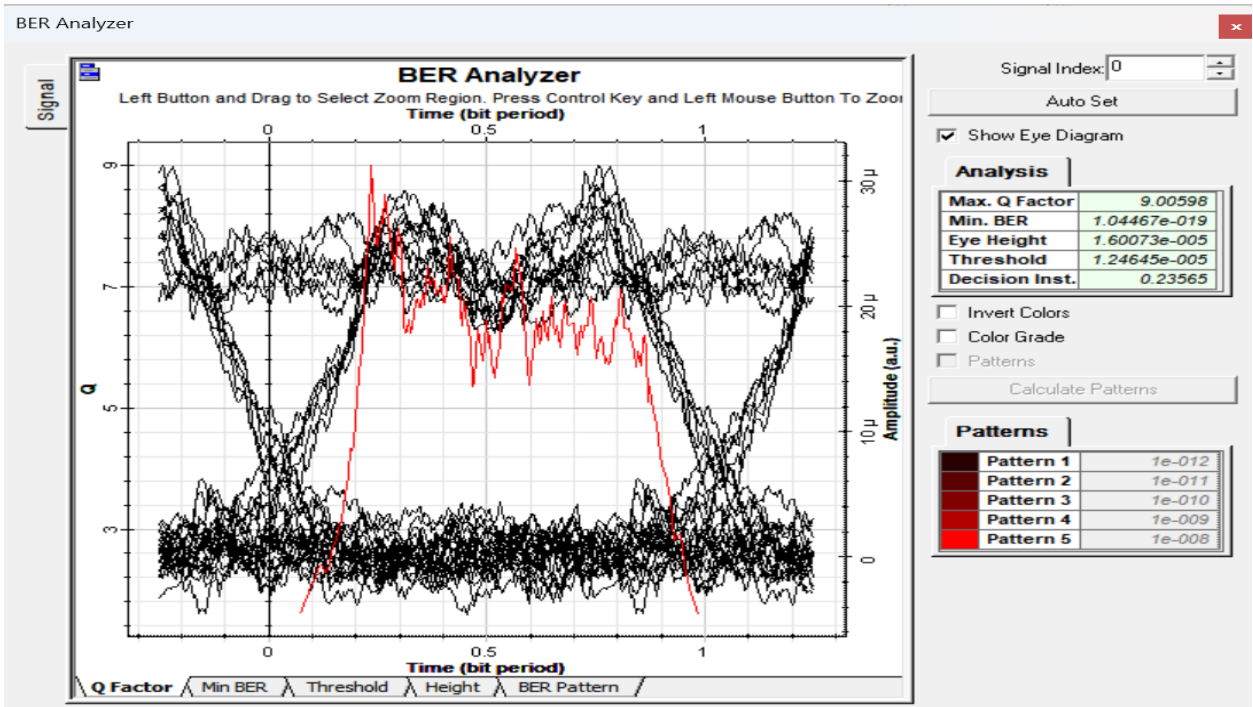


Figure 6.2: BER Analyzer — Eye Diagram After

4.5.1 Bit Error Rate (BER)

Before Optimization:

The BER Analyzer reports a Min BER of 1 before optimization — the worst possible value, indicating that every received bit is decoded incorrectly. The completely closed eye diagram with zero eye height confirms that the signal is entirely unrecoverable under the degraded system conditions.

After Optimization:

The BER Analyzer reports a minimum BER of 1.04467×10^{-19} , significantly below the ITU-T G.826 error-free threshold of 10^{-12} . This confirms fully error-free operation without Forward Error Correction (FEC), representing a complete recovery from total system failure to outstanding transmission performance. The improvement from BER = 1 to BER = 1.04467×10^{-19} demonstrates the critical impact of correct system parameter configuration on link performance.

4.5.2 Q Factor

Before Optimization:

The BER Analyzer reports a Q Factor of 0 before optimization, confirming that there is absolutely no measurable separation between the '1' and '0' signal levels at the decision point. This corresponds directly to the BER of 1 — the system cannot distinguish between logic levels at all.

After Optimization:

The BER Analyzer reports a maximum Q Factor of 9.00598 (approximately 19.09 dB), comfortably exceeding the standard error-free threshold of Q = 6.0 (15.6 dB) by a margin of 3.0. The improvement from Q = 0 to Q = 9.00598 demonstrates the critical importance of correct system parameter configuration — specifically the 30 dB extinction ratio MZM, 100 km fiber with 0.15 dB/km attenuation, order-2 Butterworth filter, and unity-responsivity PIN detector working together to deliver excellent signal quality at the receiver decision point.

5. Conclusion

This report presented the simulation and performance analysis of an optical fiber communication system using NRZ-OOK modulation in OptiSystem, evaluated under both degraded and optimised configurations. The link comprised a PRBS Generator, NRZ Pulse Generator, CW Laser (193.1 THz, 0 dBm), Mach-Zehnder Modulator (30 dB extinction ratio), 100 km single-mode fiber (0.15 dB/km), PIN Photodetector (1 A/W), and second-order Low Pass Butterworth Filter ($0.75 \times \text{Bit rate}$).

Before optimisation, the system experienced complete failure — the transmitted signal showed no visible pulse pattern, the received signal was entirely noise-dominated, and the BER Analyzer reported Q Factor = 0 and Min BER = 1, confirming that no meaningful data recovery was possible under the degraded conditions.

After optimisation, all performance metrics improved dramatically. The received optical power measured -4.554 dBm at the fiber output, and the second-order Butterworth filter effectively suppressed out-of-band noise with a controlled 1.404 dB power reduction. The optical spectrum maintained a clean carrier peak at $1.55252 \mu\text{m}$ with a well-resolved sinc-squared envelope, confirming linear-regime propagation throughout the 100 km span. Most significantly, the BER Analyzer reported a Q Factor of 9.00598 (19.09 dB) and a minimum BER of 1.04467×10^{-19} — far exceeding the ITU-T G.826 error-free threshold of $Q \geq 6.0$ and $\text{BER} \leq 10^{-12}$ — with a clearly open eye diagram confirming reliable signal detection.

The results demonstrate that correct parameter configuration is critical to achieving error-free optical transmission. Future work could explore extended reach using Erbium-Doped Fiber Amplifiers (EDFAs), dispersion compensation for higher bit rates, and advanced modulation formats such as DPSK or coherent QAM for improved spectral efficiency.